

No.

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Ex. 2.22 | Ch. 2 | P. 60 | Chloe | 2025. 7. 20

时: 根据狄拉克 δ 函数的筛选性, 若 $f(x)$ 在 $x=a$ 处连续, 则

$$\int_{-\infty}^{+\infty} f(x) \delta(x-a) dx = f(a) \int_{-\infty}^{+\infty} \delta(x-a) dx = f(a)$$

$$(a) \int_{-3}^{+1} (x^3 - 3x^2 + 2x - 1) \delta(x+2) dx$$

$$= [(-2)^3 - 3(-2)^2 + 2(-2) - 1] \int_{-3}^{+1} \delta(x+2) dx \quad (x+2=a, a=-2 \in (-3, 1))$$

$$= -25$$

$$(b) \int_0^{+\infty} [\cos(3x) + 2] \delta(x-\pi) dx$$

$$= (\cos 3\pi + 2) \int_0^{+\infty} \delta(x-\pi) dx = 1$$

$$(c) \int_{-1}^{+1} e^{(|x|+3)} \delta(x-2) dx$$

$$x-2 \Rightarrow a=2 \notin [-1, 1], \text{ 故积分为 } 0$$

Ex. 2.26 / Ch. 2 / P60 / Chloé / 2025.T.20

证: 要证明 $\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} dk$

利用傅里叶变换

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(k) e^{ikx} dk \Leftrightarrow F(k) = \int_{-\infty}^{+\infty} f(x) e^{-ikx} dx$$

$$\text{取 } f(x) = \delta(x), F(k) = \bar{\delta}(k)$$

$$\bar{\delta}(k) = \int_{-\infty}^{+\infty} \delta(x) e^{-ikx} dx = e^{-ik \cdot 0} = 1$$

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} dk$$

得证